AWS VPC Peering with Bastion Host and Private EC2 Instances

EWB Student Project Report

Submitted by	Piyush Singh
Cloud region	Asia Pacific (Sydney) — ap-southeast-2
Project scope	Two VPCs, VPC Peering, route tables, security groups, Bastion SSH, private connectivity and ICMP failure/recovery testing
Evidence repository	github.com/PiyushSingh005-AMM/aws-vpc-peering-project
Report date	24 October 2026

This report documents the AWS configuration and test evidence contained in the submitted project ZIP. Resource identifiers and test results are reproduced from the project notes and screenshots; any item not evidenced is explicitly marked for confirmation.

1. Project objective

Build private IP connectivity between two Amazon VPCs using VPC Peering, configure routing in both directions, and use a Bastion Host as the SSH entry point to private EC2 instances. Validate connectivity with ICMP tests and demonstrate how a security-group rule can block and restore traffic.

2. Architecture and resource inventory

The deployment is in AWS Region **ap-southeast-2 (Sydney)**. VPC-A contains a public subnet for the Bastion Host and a private subnet for App-A. VPC-B contains a private subnet for App-B. The two VPCs are connected through an active peering connection.

Resource	Identifier / CIDR / address
VPC-A	vpc-0295c8550008c1ec7 — 10.10.0.0/16
VPC-B	vpc-02e5cc8f332d7c490 — 10.20.0.0/16
VPC Peering	pcx-0f8e4d0dd06363ea0 — Peer-VPC-A-to-B — Active
Public subnet A	subnet-068f9d57fdd495da3 — 10.10.1.0/24
Private subnet A	subnet-0be8d4fe565814d38 — 10.10.2.0/24
Private subnet B	subnet-0ae5afb6dc4f8d34d — 10.20.1.0/24
Bastion Host	Private IP 10.10.1.10; public IP 3.25.174.208
App-A	Private IP 10.10.2.10; no public IPv4 shown in project notes
App-B	Private IP 10.20.1.10; no public IPv4 shown in project notes

Note: Instance IDs for App-A and App-B are not included in the supplied project notes, so they are not guessed in this report.

3. Route tables and associations

The following route entries are documented in the repository README/project notes. The local route in each VPC enables intra-VPC routing. The peering route directs traffic for the remote VPC CIDR to the peering connection. The public route table also includes a default route to the Internet Gateway for the public subnet.

Route table	Associated subnet	Destination	Target
RT-A-Public (rtb-01ba4d269db315311)	Subnet-A-Public	10.10.0.0/16	local
RT-A-Public	Subnet-A-Public	10.20.0.0/16	pcx-0f8e4d0dd06363ea0
RT-A-Public	Subnet-A-Public	0.0.0.0/0	Internet Gateway
RT-A-Private (rtb-050ba6b9c1888ad52)	Subnet-A-Private	10.10.0.0/16	local
RT-A-Private	Subnet-A-Private	10.20.0.0/16	pcx-0f8e4d0dd06363ea0
RT-B-Private (rtb-014169493e2761079)	Subnet-B-Private	10.20.0.0/16	local
RT-B-Private	Subnet-B-Private	10.10.0.0/16	pcx-0f8e4d0dd06363ea0

4. Security groups

SG-Bastion (sg-091d6bf50a23d6222): SSH (TCP 22) is restricted to the administrator's current public IPv4 /32, as recorded in the project setup notes.

ewb-sg-app-a (sg-0dd7c4f424b3200da): SSH (TCP 22) from 10.10.1.10/32 (Bastion private IP); ICMP Echo rule used for testing traffic from App-B as configured during troubleshooting.

ewb-sg-app-b (sg-06e24dc8976a02415): SSH (TCP 22) from 10.10.1.10/32 (Bastion private IP); ICMP Echo rule from 10.10.2.10/32 for App-A-to-App-B testing. The supplied README notes that a broader temporary ICMP rule was used during reverse-direction troubleshooting; review the live rule set and restrict it to the minimum required source before production use.

Security groups are stateful. For a ping test to succeed, the destination must permit inbound ICMP Echo Request from the source, and the return traffic must be allowed by the relevant rules and network routes.

5. Connectivity and SSH test results

The supplied terminal screenshots show the following test evidence:

App-A → App-B: ping to 10.20.1.10 — 4 packets received, 0% packet loss (screenshots 08 and app-a-to-app-b-ping-success).

App-B → App-A: ping to 10.10.2.10 — 4 packets received, 0% packet loss (screenshots 09 and app-b-to-app-a-ping-success).

Private SSH: Bastion access was verified, and App-B was reached by its private address 10.20.1.10. The private SSH screenshot shows the Amazon Linux 2023 login banner and hostname for App-B. App-A was also accessed from the Bastion according to the project notes.

ICMP blocked/recovered: The repository contains screenshots named 11-icmp-blocked.png and 12-icmp-restored.png (and duplicate-name variants). The blocked screenshot appears to show a failed ping with 100% packet loss, and the restored screenshot appears to show successful replies. The precise direction/rule removed should be checked against the screenshot terminal commands and AWS security-group rule before final submission; the README currently describes the reverse direction, while the earlier project notes describe removing App-B's inbound ICMP rule for App-A → App-B. This report does not silently reconcile that discrepancy.

6. How the traffic flows

App-A sends packets to 10.20.1.10. The route table associated with App-A's private subnet forwards destination 10.20.0.0/16 to the VPC Peering connection. VPC-B's private route table has the reciprocal route for 10.10.0.0/16, allowing return traffic. The security groups must allow the relevant ICMP traffic. SSH administration is performed through the Bastion Host at 10.10.1.10, avoiding the need for public IPv4 addresses on App-A or App-B.

7. Key learning outcomes

- Created and inspected VPCs and subnets.
- Configured route tables and VPC Peering for private inter-VPC routing.
- Used a Bastion Host to reach private EC2 instances over SSH.
- Validated reachability with ping and interpreted packet loss.
- Observed how security-group ICMP rules affect connectivity and recovery.

8. Screenshot evidence

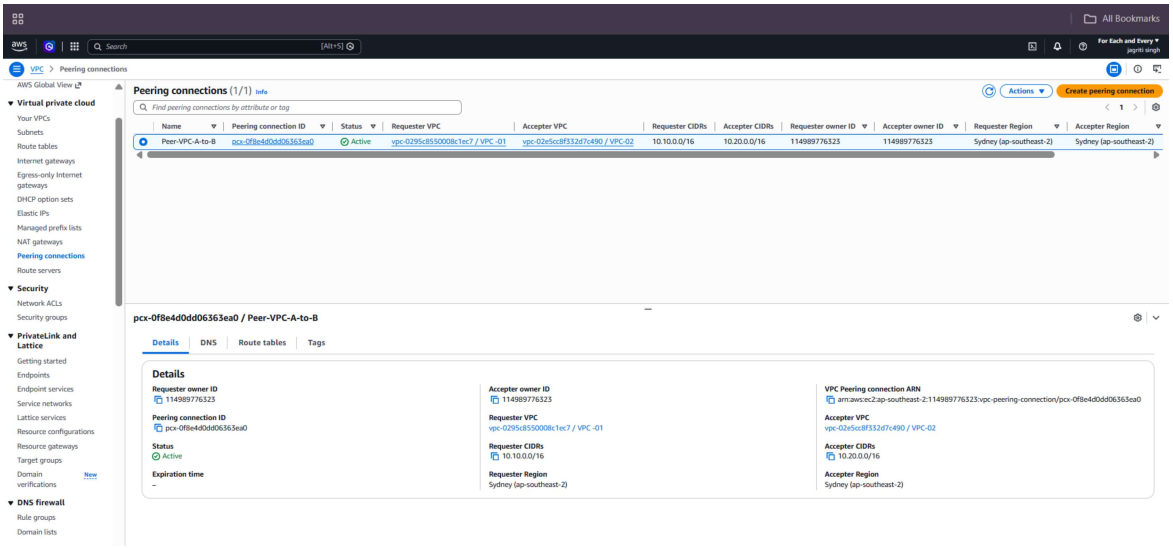


Figure 1. VPC inventory and CIDR configuration. (01-vpcs.png)

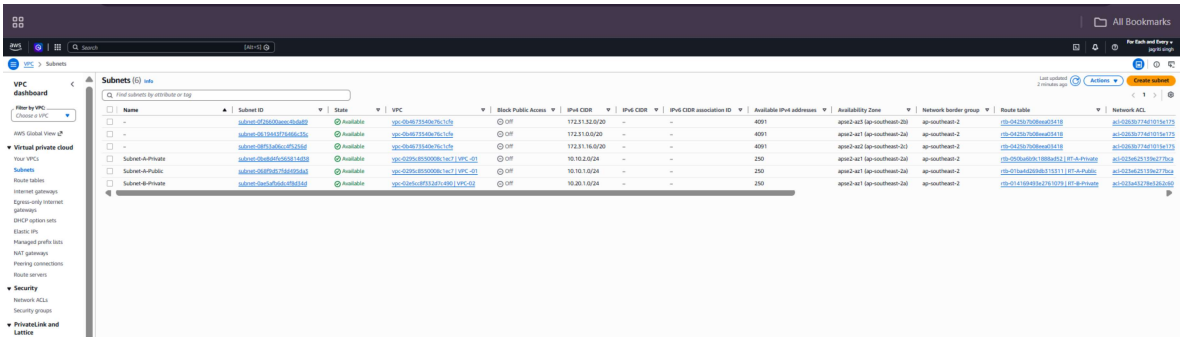


Figure 2. Subnets created for VPC-A and VPC-B. (02-subnets.png)

8. Screenshot evidence (continued)

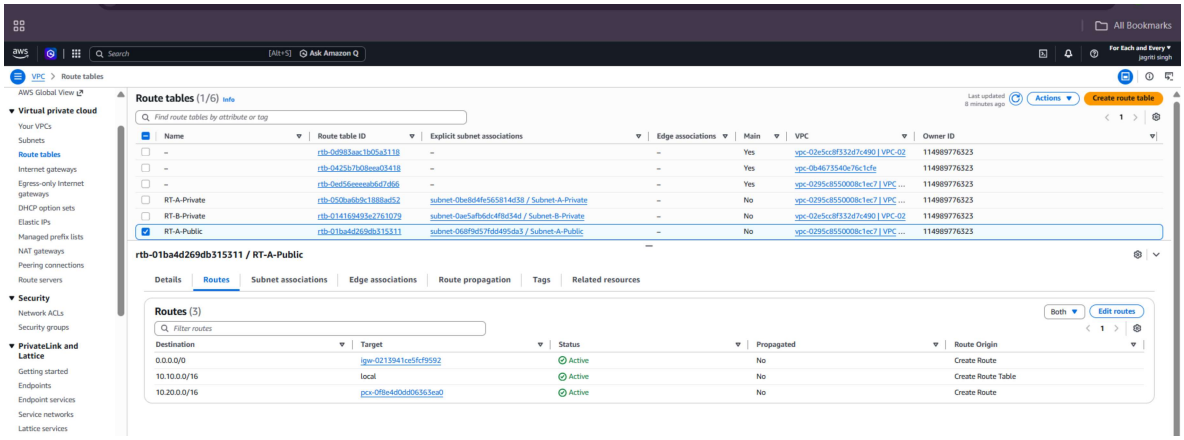


Figure 3. Public route table: local, peering and Internet Gateway routes. (04a-public-routes.png)

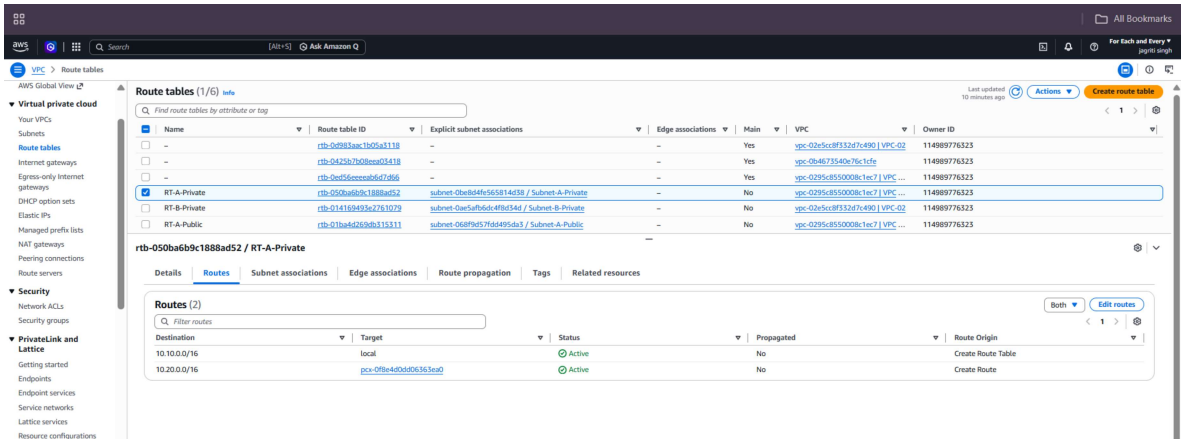


Figure 4. VPC-A private route table and peering route. (04b-private-a-routes.png)

8. Screenshot evidence (continued)

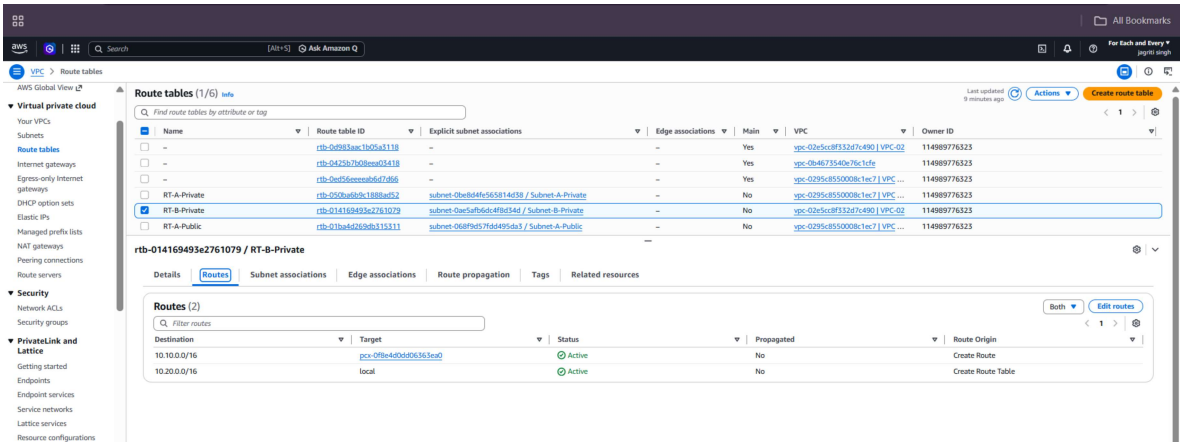


Figure 5. VPC-B private route table and reciprocal peering route. (04c-private-b-routes.png)

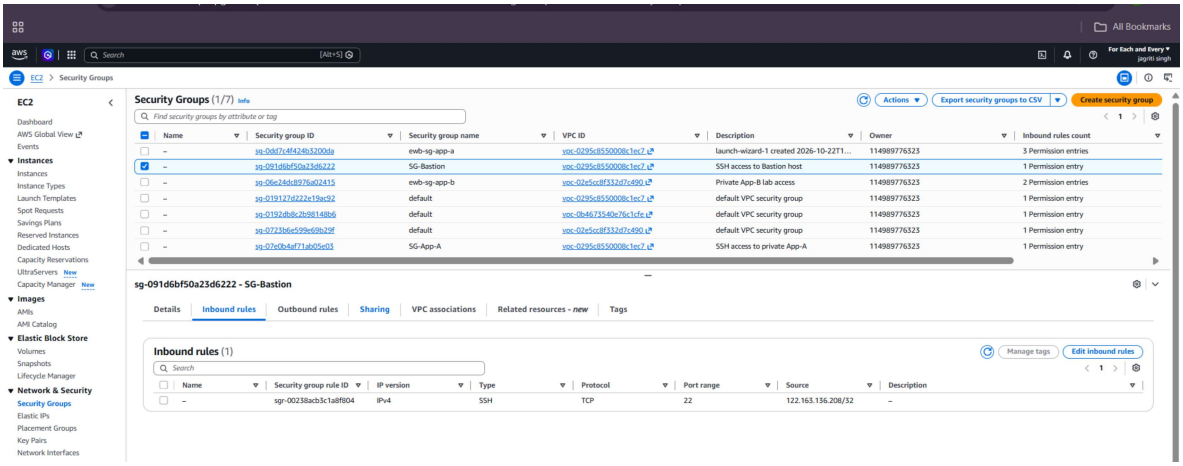


Figure 6. Bastion security-group configuration. (05a-bastion-sg.png)

8. Screenshot evidence (continued)

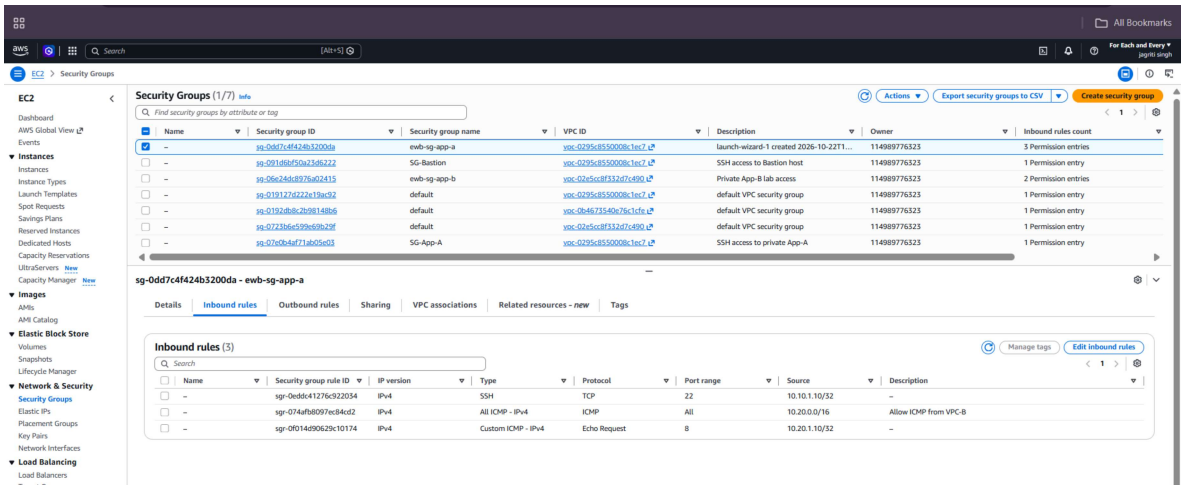


Figure 7. App-A security-group configuration. (05b-app-a-sg.png)

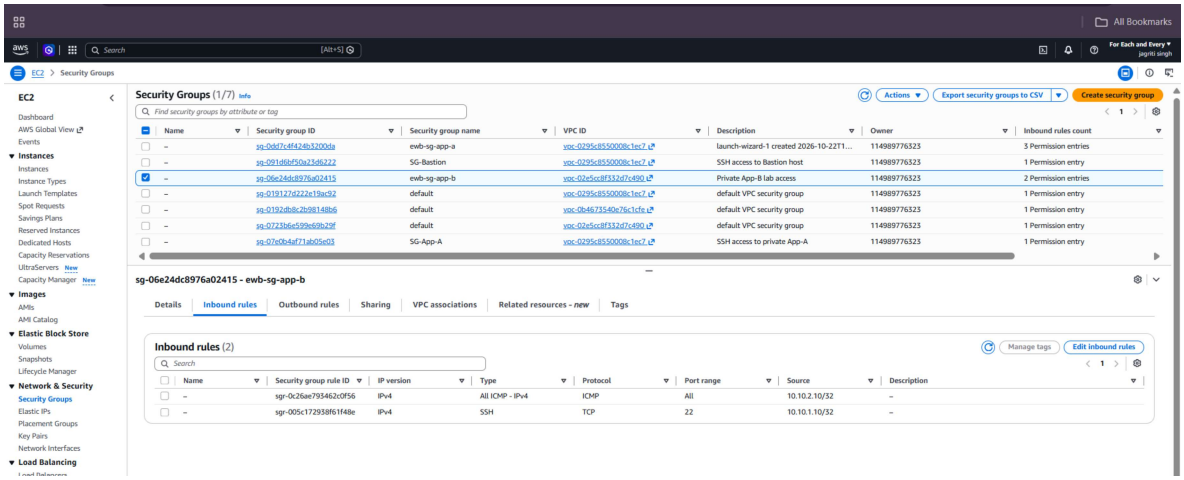


Figure 8. App-B security-group configuration. (05c-app-b-sg.png)

8. Screenshot evidence (continued)

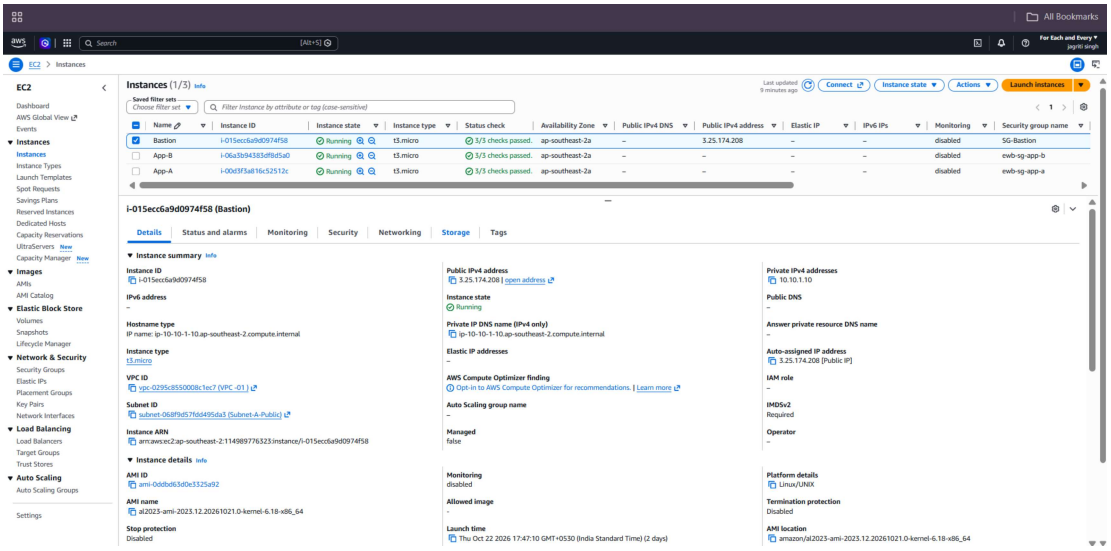


Figure 9. Bastion EC2 instance details. (06a-bastion.png)

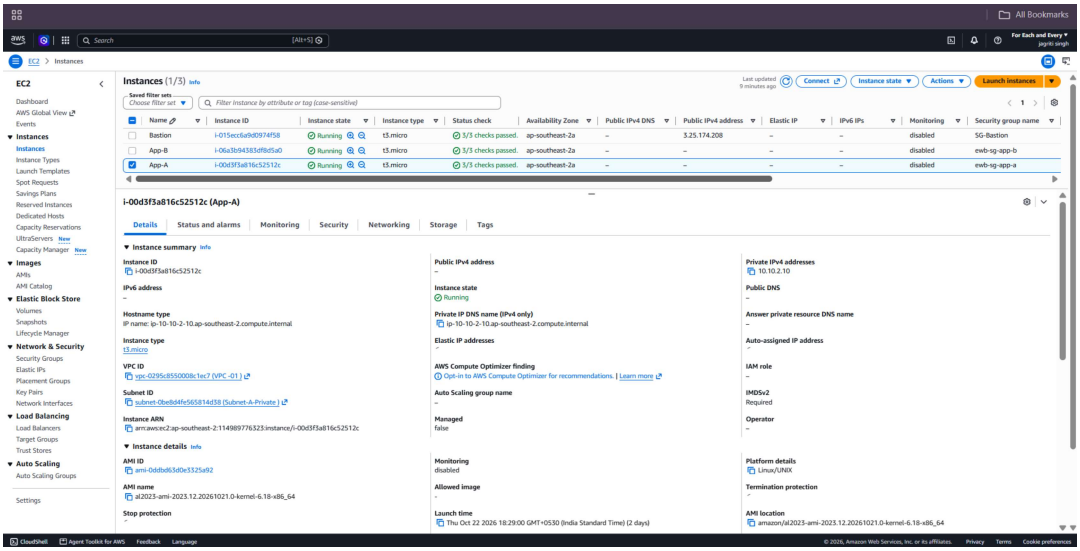


Figure 10. App-A EC2 instance details. (06b-app-a.png)

8. Screenshot evidence (continued)

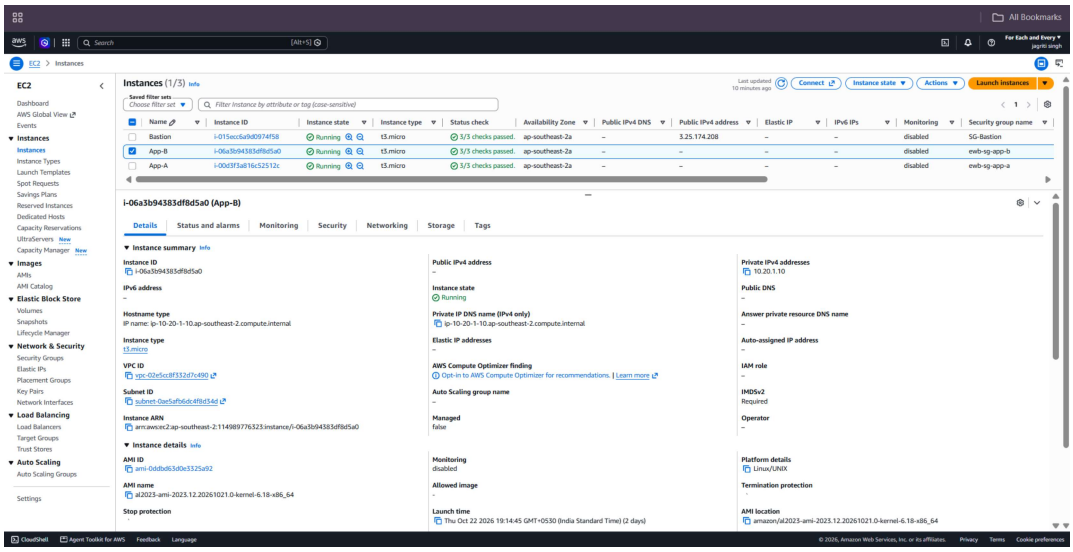


Figure 11. App-B EC2 instance details. (06c-app-b.png)



Figure 12. SSH session to the Bastion Host. (07-bastion-ssh.png)

8. Screenshot evidence (continued)

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.180 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.177 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.160 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.191 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3131ms
rtt min/avg/max/mdev = 0.160/0.177/0.191/0.011 ms
```

Figure 13. Successful ping from App-A to App-B. (08-app-a-to-app-b.png)

```
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.163 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.159 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.168 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.170 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3111ms
rtt min/avg/max/mdev = 0.159/0.165/0.170/0.004 ms
```

Figure 14. Successful ping from App-B to App-A. (09-app-b-to-app-a.png)

8. Screenshot evidence (continued)

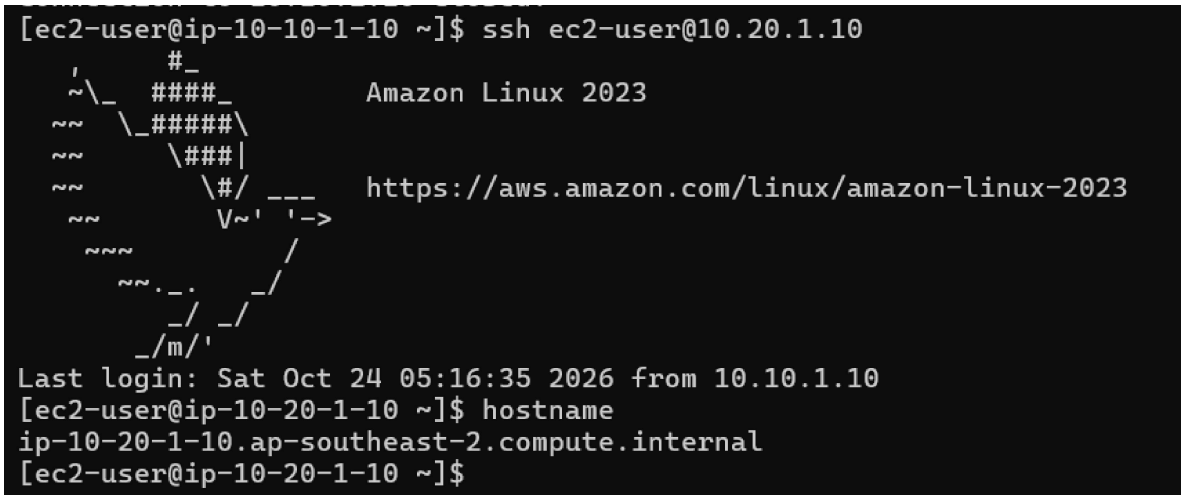


Figure 15. SSH to App-B using its private IP. (10-private-ssh-app-b.png)

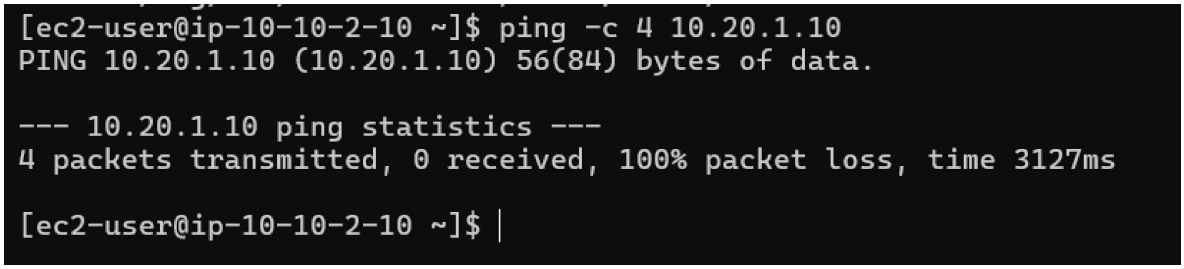


Figure 16. ICMP blocked test evidence. (11-icmp-blocked.png)

8. Screenshot evidence (continued)

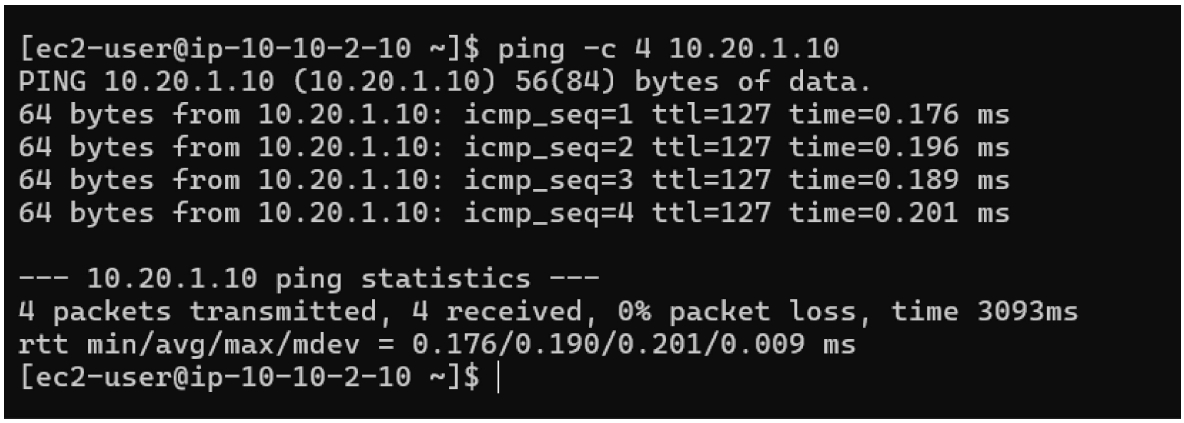


Figure 17. ICMP restored test evidence. (12-icmp-restored.png)

9. Submission checklist and limitations

The repository ZIP contains VPC, subnet, route-table, security-group, instance, SSH, bidirectional ping, and ICMP block/recovery screenshots. This PDF embeds the principal screenshots as visual evidence.

Before submitting to EWB, verify these items:

1. Confirm your EWB batch/cohort name and add it to the cover if required by your coordinator.
2. Confirm the ICMP blocked test's exact direction and the precise security-group rule changed. Keep the matching screenshot and README description consistent.
3. If the EWB rubric requires a baseline ping before peering, include that baseline failure screenshot and state the test timing accurately. It is not clearly established in the supplied notes whether a pre-peering baseline screenshot is included.
4. Add a cleanup screenshot or confirmation only after you have actually terminated lab resources and removed unused networking resources. Cleanup is not confirmed by the supplied ZIP.
5. Confirm the final report filename matches the exact submission instruction from your EWB course.

10. Cleanup and cost control

After all evidence has been saved and reviewed, terminate the three lab EC2 instances, then inspect and delete lab-only resources in dependency order: peering connection and peering routes, security groups (after no longer attached), subnets, custom route tables, Internet Gateway (detach first), and the two VPCs. Check for leftover EBS volumes/snapshots and Elastic IPs, and review AWS Billing/Cost Explorer after usage reporting updates. Do not claim cleanup is complete until each action has been verified.

Evidence repository: https://github.com/PiyushSingh005-AMM/aws-vpc-peering-project
Screenshots folder: https://github.com/PiyushSingh005-AMM/aws-vpc-peering-project/tree/main/screenshots

End of report